GUEST LECTURE · ELECTRONICS & INSTRUMENTATION ENGINEERING · VRSEC

AI and Digital Twins in Instrumentation

SHAFT f_r

29.95 Hz

OUTER RACE BPFO

107.4 Hz

VISIBLE IN A RAW FFT

NO

What they actually do, what they aren't allowed to do, and where the trap is.

THE PREMISE

Instrumentation was never limited by our ability to measure. It was limited by what we could do with the measurement.

That second half is the part that changed. That is where AI lives.

Who is talking, and how to discount him

WHAT I HAVE DONE

- Graduate of this department
- 13 years consulting — TCS, Cognizant, Accenture
- Enterprise document platforms for banks and insurers
- Data pipelines and governance in a regulated industry

WHAT I HAVE NOT DONE

- Never commissioned a control loop
- Never signed off a safety instrumented system
- Never worked a plant turnaround
- If you work in automation, correct me in the Q&A

The model is never why these projects fail. The data pipeline, the failure modes and the trust are. Those lessons transfer completely.

Five parts

	The stack, honestly	8 min	Where AI sits, and where it is not allowed to sit
	One failure, all the way down	18 min	A single bearing — geometry to signal processing to a decision
	Demo one: Bearing Fault Lab	6 min	Take one measurement apart
4	**Digital twins**	11 min	The taxonomy, the trap, and demo two — held to its own test
	What to learn, what the jobs are	6 min	Including which of your electives are worth picking

Where this sits in your syllabus

YOU ALREADY HAVE THIS

- Python — Semester II, core
- FFT and filter design — DSP, Sem VI
- Sampling and aliasing — DSP
- PID and loops — Control Systems
- PLC, DCS, SCADA — Industrial Automation
- MPC — Computer Control of Processes

NOT IN YOUR SYLLABUS AT ALL

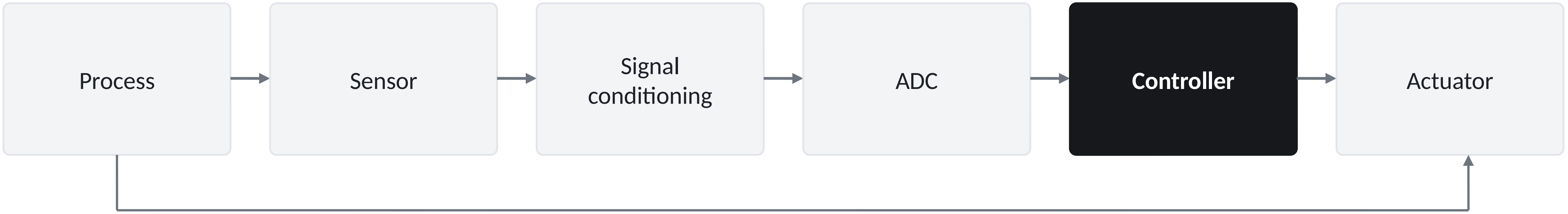
- Hilbert transform and envelope detection
- Probability and statistics — no course exists
- Kurtosis and higher-order statistics
- Vibration analysis and condition monitoring
- Modbus, OPC UA, MQTT
- Digital twins, anomaly detection

ELECTIVE ONLY — WHO TOOK IT?

- IEC 61508 → Safety Instrumentation Systems
- Historians, IIoT data flow → Industrial IoT
- Machine learning → AI/ML in Healthcare
- HART, PROFIBUS → Industrial Comm. Networks

Everything in the middle column is why this talk is worth an hour.

The chain has not changed



WHY 4-20 mA IS STILL 4-20 mA

Current, not voltage, because current does not care about lead resistance. Live zero at 4 mA so you can tell "reading zero" apart from "the wire is broken." That was good engineering in 1955 and it is good engineering now. HART still rides digital data on top of it. AI does not delete any of this — it sits on top of it.

Where the layers sit

4 / 5	**Enterprise IT**	ERP, cloud, data lake	minutes to hours
3	**Site operations**	Historian, MES, batch records	seconds to minutes
2	**Supervisory**	SCADA, HMI — operators look here	sub-second
1	**Basic control**	PLC / DCS — PID on a deterministic scan	10–100 ms
0	**The field**	Sensors and actuators — physics	continuous

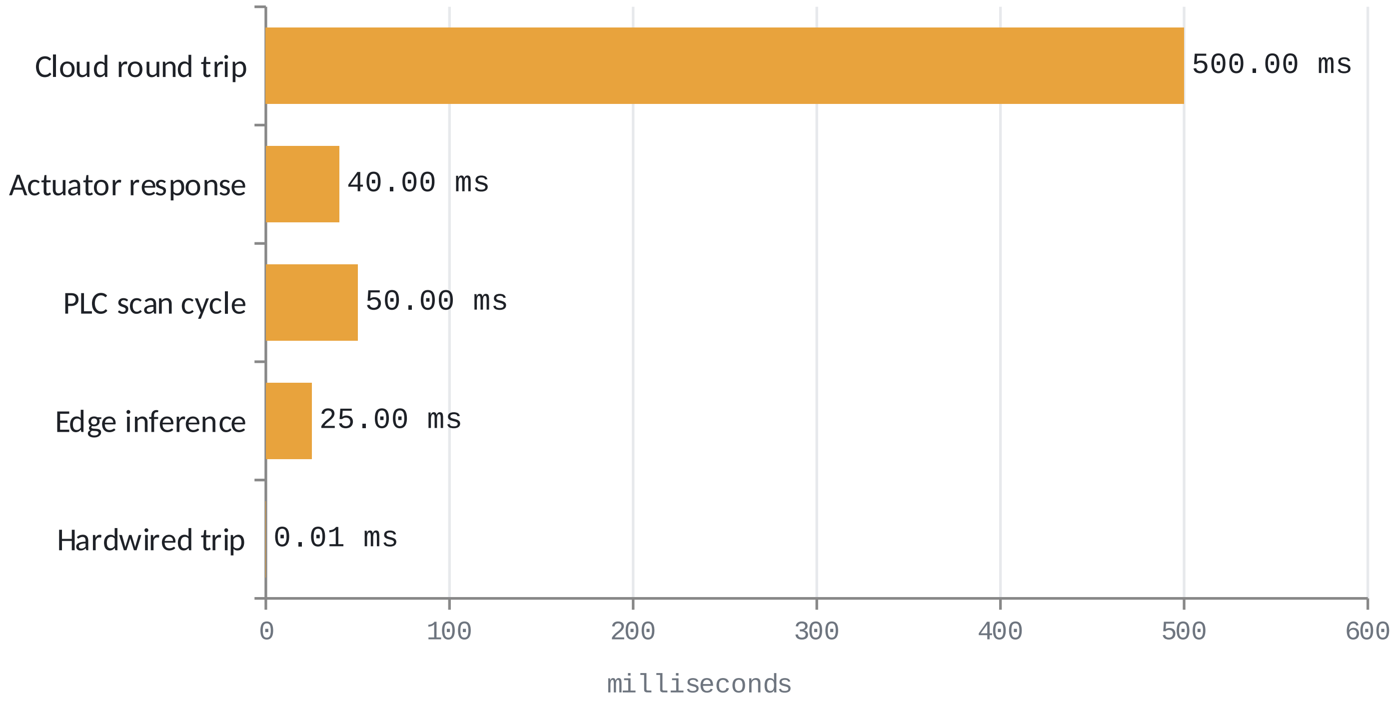
Going up trades latency for context — and data flows up far more freely than commands flow down. That asymmetry is a security control.

How the data actually moves

4-20 mA / HART	Field device to controller. Analog loop with digital riding on top.
Modbus	Ancient, trivially simple, absolutely everywhere. No security whatsoever.
PROFINET / EtherNet/IP	Industrial fieldbuses on Ethernet.
OPC UA	The important one. Carries meaning, not just numbers — a tag knows it is a temperature in °C on a named asset.
MQTT + Sparkplug B	Lightweight publish/subscribe. How edge devices report upward.

Then it lands in a historian: AVEVA PI or AspenTech IP.21 in process industries; InfluxDB or TimescaleDB in newer stacks. If a vendor cannot tell you how their product gets data out of your DCS, they do not have a product — 90% of the work is here, at the boring integration layer.

The microsecond claim is wrong by 10,000×



AI IN THE LOOP, REALISTIC

10–50 ms

HARDWIRED TRIP

<0.1 ms

Microsecond response belongs to hardwired interlocks and analog trip amplifiers. No software in the path — which is exactly why they are trusted with safety.

Cloud round trip is off the chart scale in practice — treat 500 ms as optimistic.

THE RULE

AI sits above the control layer, not inside it.

It advises a human or a supervisory system. It rarely has direct authority over an actuator. It essentially never has authority over a safety function. We will come back to why.

One failure, all the way down

40–50%

of failures in electrical rotating machines are bearing failures

WHY THIS EXAMPLE, FOR EIGHTEEN MINUTES

- It is not a toy — it is the single most common failure mode
- The cleanest path I know from physics → signal processing → machine learning → a business decision
- Everything is publicly verifiable: the bearing, the dataset, the arithmetic
- And it shows exactly where the AI is, and how small that part is

The physics gives you the answer first

THE CONSTRAINT

Balls roll rather than slide. That single constraint fixes the cage speed as a fraction of shaft speed. Everything else follows by counting how often a ball passes a defect.

n = ball count d = ball diameter
D = pitch diameter θ = contact angle

FTF

$$(f_r / 2) \cdot (1 - (d/D)\cdot\cos\theta)$$

cage rotation

BPFO

$$(n/2) \cdot f_r \cdot (1 - (d/D)\cdot\cos\theta)$$

ball passes a fixed point on the outer race

BPFI

$$(n/2) \cdot f_r \cdot (1 + (d/D)\cdot\cos\theta)$$

...on the rotating inner race

SKF 6205 at 1797 rpm

9 balls · $d = 7.94$ mm · $D = 39.04$ mm · $\theta = 0$ · $d/D = 0.2034$

SHAFT f_r	CAGE FTF	OUTER BPFO	INNER BPFI
29.95 Hz	11.93 Hz	107.4 Hz	162.2 Hz

We know the frequency the fault will announce itself at — before measuring anything.

That is not machine learning. It is mechanics, and it is worth more than any model, because it turns an open-ended search into looking in one specific place.

One design detail worth noticing: BPFO is 3.585× shaft speed. Not 3, not 4. Manufacturers choose ball counts so defect frequencies are deliberately non-integer multiples — so faults cannot hide underneath shaft harmonics.

So why can't we just take an FFT?

The fault is tiny

An early spall produces impacts of very low energy. In the demo, the 1× shaft line is about 170× larger than anything the defect puts at 107 Hz.

The fault is broadband

A very short impact is not narrowband. It does not deposit energy at 107 Hz — it rings the housing resonance, typically 2–15 kHz, which then decays.

The information is the rate

What carries the diagnosis is not the ringing frequency. It is how often the ringing repeats.

This is amplitude modulation: the defect modulates, the resonance carries. Demodulation is not in your syllabus — so we are going to build it from scratch, and it takes four minutes.

Demodulation in four minutes

Take the DFT

The transform you compute with the FFT in DSP. For a real signal the spectrum is symmetric — the negative half mirrors the positive half and carries nothing new.

Delete the negative frequencies

Zero them. Double the positive ones so the energy still balances.

Transform back

What returns is complex-valued. This is the analytic signal, and the operation that produced it is the Hilbert transform.

4 **Take the magnitude**

Real squared plus imaginary squared, square rooted. The oscillation cancels and the outline survives. That outline is the envelope.

Why the magnitude kills the wiggle: if you have sin(t) and cos(t), the magnitude is exactly 1, forever. You have traded an oscillation for its amplitude.

Envelope analysis

	Band-pass the resonance	2.4–4.0 kHz. Throwing away the low end deletes the shaft imbalance that was drowning everything.
	Take the analytic signal	Hilbert transform, then magnitude. Carrier stripped, envelope left.
	Remove DC, then FFT again	Transform the envelope itself, not the original signal.
4	**Look at 107.4 Hz**	A clean line at BPFO, with harmonics. Exactly where geometry said.

We used domain knowledge to build a feature in which the fault is obvious. Throw raw vibration at a deep network instead and you are asking it to rediscover the Hilbert transform and bearing kinematics from data. It might. It will need vastly more data, and you will not be able to explain what it learned.

Now, finally, the machine learning

Notice how much smaller this section is than the signal processing. That ratio is correct.

FEATURES

- RMS — total energy, rises late
- Kurtosis — 4th moment, measures impulsiveness, rises early
- Crest factor — peak / RMS
- Envelope band energy at BPFO and harmonics

FRAMING A — SUPERVISED RUL

"This bearing has 340 hours left."

Needs run-to-failure data: machines deliberately instrumented and allowed to fail, repeatedly, across every mode you care about.

Ask how many companies have destroyed a statistically useful number of their own machines.

Essentially none. Which is why the same four public datasets appear in every paper.

FRAMING B — ANOMALY DETECTION

Train only on the healthy machine. Learn normal. Alarm on deviation.

Autoencoder reconstruction error, one-class SVM, isolation forest — or honestly, a control chart on kurtosis.

This is what actually gets deployed, because healthy data is the only thing you have a lot of.

Why these projects fail

You have no labels	Not few. None. A thousand motors over five years might give you a dozen documented failures with usable data.
Models do not transfer	Different mounting stiffness moves the resonance. Domain shift is brutal. Note the physics transfers perfectly — BPFO is BPFO.
Sensor drift	Sensitivity shifts with temperature and age. A large share of anomaly alerts are instrumentation faults. Monitor your monitoring.
Alarm fatigue	The one that actually kills deployments. Next slide does the arithmetic.
Nobody set the threshold economically	It was set by optimizing F1 score instead of by costing out consequences.

The success stories are all in the vendor brochures. The failures are where the engineering is.

Do the arithmetic. Watch the number.

ASSETS MONITORED	FALSE POSITIVE RATE	FALSE ALARMS PER DAY	INVESTIGATION LOAD
500	1 %/day	5	15 hrs/day

THE BUDGET YOU ARE SPENDING

EEMUA 191 guidance: an operator can absorb roughly one alarm every ten minutes in steady state — across all alarm sources. Your analytics system competes for that budget with every process alarm in the plant.

WHAT ACTUALLY HAPPENS

By week three the engineers have learned the system cries wolf, and stop opening tickets. Six months later a real failure is flagged, ignored, and the machine fails.

The postmortem says the model was not accurate enough. The model was fine.

The threshold was set by a data scientist optimizing F1 instead of by an engineer costing out consequences.

Set the threshold with a cost model

```
P*  =  C_FP / (C_FP + C_FN)
```

Alarm when the expected cost of not acting exceeds the cost of acting. C_FP is an unnecessary inspection; C_FN is an unplanned failure.

INSPECTION COST C_FP	UNPLANNED FAILURE C_FN	OPTIMAL THRESHOLD P*
$2k	$200k	1 %

So you should alarm at 1% confidence and accept many false positives — but only if you have the inspection capacity to absorb them. If you do not, the binding constraint is how many investigations your team can run per week. The threshold is a capacity planning decision, not a modelling decision.

Where AI goes in control (mostly: not in the loop)

THE WORKHORSE — MPC

Model predictive control has been standard in refineries since the 1980s. It solves a constrained optimization at every step over a prediction horizon.

It handles what PID cannot: interacting variables, hard constraints, dead time. It is not machine learning — though its internal model can be learned.

REINFORCEMENT LEARNING — BE CAREFUL

RL learns by trying things. In a live process, "trying things" means potentially driving a reactor somewhere it should not go.

The well-known success is data centre cooling — and note how: advisory mode first, hard constraints, operator override at any time. A supervisory optimizer choosing setpoints, not a controller moving valves.

Honest state of the art: RL sets targets. Conventional control hits them.

IF YOU REMEMBER ONE THING FROM THIS HOUR

You cannot assign a SIL rating to a neural network.

IEC 61508 and 61511 govern safety functions. A Safety Integrity Level is a claim about probability of dangerous failure on demand — SIL 3 means between 1 in 1,000 and 1 in 10,000. To claim it you must demonstrate that rate with evidence, across the whole lifecycle.

You cannot enumerate its failure modes	You cannot bound behaviour on untested inputs	Its behaviour changes when you retrain it

Not AI pessimism. This is what mature safety engineering looks like — and the AI industry is slowly rediscovering it under the name "guardrails." You already have a discipline that solved this. Do not let anyone talk you out of it.

Bearing Fault Lab

kiranic.com/lecture/bearing-lab.html

One measurement, taken apart. Raise the defect and watch where it shows up: barely in the raw spectrum, unmistakably in the envelope spectrum.

HEALTHY — BPFO INDEX

0.3

EARLY DEFECT — ALERT

3.1

ADVANCED — FAULT

7.6

SAME DEFECT, NOISY PLANT

1.6

Noisy plant — the false negative. Same 22% defect, nothing changed about the bearing — the index falls to 1.6 and the annunciator returns to NORMAL. The fault is missed, and nobody is told. A false alarm is annoying and visible; a missed detection is silent.

What a digital twin actually is

A model of a physical asset, kept current by that asset's own data, that you can ask questions of. Not current → it is a simulation. Cannot be questioned → it is a dashboard.

1

Physics model

First principles — differential equations, FEA, kinematics. Accurate exactly where your equations are right. Slow, and it needs someone who understands the machine.

2

Data-driven surrogate

Trained to mimic the plant from its history. Fast. Interpolates well. Extrapolates terribly — hold that thought.

3

Hybrid

Physics model whose parameters are continuously re-estimated from live data. The interesting one, and by far the hardest to build.

You already built one this morning. When we derived BPFO from ball count and pitch diameter, we built a physics model of a bearing that predicts what the sensor will see. That is a twin component.

What does your twin predict that a human could not — and how was that validated?

No answer to the first half

It is a visualization.

No answer to the second half

It is a guess with good graphics.

A 3D model with live values painted on it fails both. It may be genuinely useful. It is not a twin — it cannot answer "what happens if I raise the feed rate ten percent."

The trap: a fault is extrapolation

Five years of historian data

Almost entirely normal operation. A plant that failed often enough to give you a rich failure dataset would be a scrapyard.

The surrogate learns the healthy region

Inside it, the model interpolates — and interpolation is what these models are genuinely good at.

A fault is the machine leaving that region

By definition. A fault is the machine doing something it has not done before. That is what the word means.

So the model is excellent everywhere except the one place you bought it for.

And your validation set is also normal. Hold out 20% of five years of healthy data, test on it, R^2 comes back 0.99 — the metric that tells you it works is computed on the data that cannot tell you it works. That is structural. No architecture fixes it.

Live Condition Monitoring

kiranic.com/lecture/live-monitor.html

This is a level-one twin. There is no training data anywhere in it. It runs the bearing model from Part 2 forward in time — ball count and pitch diameter give BPFO, impacts ring the housing resonance, the spall grows — and computes what the accelerometer would see.

WHAT IT PREDICTS A HUMAN COULD NOT

Days-to-fault for MTR-402. A falsifiable claim about the future — you wait, and you see.

HOW WELL IT DOES

Confidently wrong early, swinging between numbers. It settles only once you could already read it off the chart.

```
Threshold 8:  0 false alarms, 22 d warning
Threshold 2:  4–5 per 30 d, 38–46 d warning
```

I left the projection crude deliberately. That is the honest state of most remaining-life numbers in the field — better you distrust them for the right reason than trust them for the wrong one.

I hit this wall in my own house

THE WALL

A Raspberry Pi at home reads light, motion, temperature and proximity sensors and drives the automation. Sensors, controller, actuators — instrumentation at domestic scale.

I am exploring a data-driven surrogate on its history. Which means years of normal and not one failure.

THE HONEST RESPONSE

Stop trying to model the fault. Model NORMAL — learn what healthy looks like in detail, from the data you actually have, which is all healthy.

Then measure distance from it, and alarm on the distance.

You do not need to have seen a failure to know that today does not look like any day you have seen before.

That is Framing B from Part 2, and this is why it wins: it matches the data that exists rather than the data you wish you had. A twin that says "this is not normal, and here is which sensor is furthest out" is worth more than one claiming to predict a failure mode it has never observed.

What to learn, in this order

1	**Statistics**	The biggest hole in your degree — no probability or statistics course exists in eight semesters. Everything about thresholds and error rates is statistics.
2	**The scientific Python stack**	You have Python from Semester II. You do not have numpy, scipy.signal, pandas or scikit-learn. A different skill.
3	**Signal processing past DSP**	Two credits is not enough, and Signals and Systems was removed. Hilbert and envelope detection are not in it.
	One time-series database	InfluxDB or TimescaleDB. Downsampling policies and retention.
	OPC UA	Zero mentions in your syllabus — I checked. Python asyncua library. A real differentiator.
	Classical ML before deep learning	Isolation forest, one-class SVM, PCA, gradient boosting. Precision and recall well enough to defend a threshold.

This list is built against VR20. I left out everything you already get.

Choose your electives deliberately

Safety Instrumentation Systems

Open Elective 2, Sem VI

IEC 61508 and the protection-layer model. The most employable elective in your list, and almost nobody picks it. Everything on my SIL slide is in this course.

Industrial Internet of Things

Program Elective 3, Sem VII

Historians, industrial protocols, the SCADA to ERP data flow. The layer above where your core courses stop.

Real World Instrumentation with Python

Program Elective 5, Sem VII

Instrument systems, simulators, data I/O. The bridge from Semester II Python to actual instrumentation work.

AI and Machine Learning in Healthcare

Open Elective 2, Sem VI

The only real ML in your syllabus. The healthcare framing is incidental; the methods transfer.

The first and last are both Open Elective 2, so you must choose. Take the safety one — safety is much harder to self-teach and much harder to fake in an interview.

A project you can start this week

Download the Case Western Reserve bearing dataset — public, the standard benchmark, and the exact 6205 bearing from today.

1. **Reproduce today's demo on real data** — Compute the envelope spectrum. Confirm the peak lands at the BPFO you calculated from geometry. When it does, you have verified theory against measurement — which is the actual job.

2. **Build a detector trained only on healthy files** — Anomaly detection, not classification. You do not have the labels for classification and neither does industry.

3. **Report your false positive rate honestly** — And argue for a threshold using a cost model you state explicitly. Almost nobody does this part.

Do step three and you can have a conversation in an interview that most candidates with a master's degree cannot have.

The scarce skill is not the model.

"AI engineer" is mostly not where these jobs are. They are: controls engineer who can code. Reliability engineer who understands statistics. Data engineer who understands process. Architect who can talk to both the plant and IT.

> The scarce skill is the person who understands the physical process AND the software. You are two years from being that person, and most of you do not realize it is valuable.

THE MISTAKE I MADE

I waited too long to learn the new tooling. I assumed depth in the old thing was enough. It was not, and the people who moved earlier had a far easier transition than I did.

Get deep in something real — EIE is something real. Then learn the tools relentlessly, starting now, not in year five. Depth plus current tooling is a very strong position. Either one alone is not.

Cheat sheet

QUANTITY	FORMULA	6205 @ 1797 rpm
Shaft	f_r = rpm / 60	29.95 Hz
Cage	FTF = (f_r/2)(1 − (d/D)cos θ)	11.93 Hz
Outer race	BPFO = (n/2)·f_r·(1 − (d/D)cos θ)	107.4 Hz
Inner race	BPFI = (n/2)·f_r·(1 + (d/D)cos θ)	162.2 Hz
Ball spin	BSF = (D/2d)·f_r·(1 − ((d/D)cos θ)²)	70.6 Hz

BSF note: a ball spall strikes both races once per ball revolution, so the observed line is usually 2 × BSF = 141.2 Hz. Published tables (including Case Western) tabulate that doubled value. Check the convention before hunting for a peak.

Standards: IEC 61508 / 61511 (SIL) · IEC 62443 (OT security) · ISO 20816 (vibration severity) · EEMUA 191 (alarms) · OPC UA = IEC 62541 · Purdue / ISA-95

ENVELOPE SIGNATURES

- Outer race — clean BPFO + harmonics, no sidebands. Easiest.
- Inner race — BPFI with ±f_r sidebands.
- Rolling element — BSF with cage-rate sidebands.

NUMBERS WORTH KEEPING

- AI in loop: tens of ms, not µs
- Bearing resonance: 2–15 kHz
- Resolution = 1 / record length
- P* ≈ C_FP / (C_FP + C_FN)

Questions.

Including the hostile ones.

Both simulators are yours. Open them, break them, read the source — every number in this talk was computed by that code, and it is about four hundred lines.

```
kiranic.com/lecture
```

Scan for both demos